

# EpiTox Paper Restructuring Guide:

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## Keeping EpiTox as Main Character for BOTH Audiences

### The Real Challenge

**Current Situation:** Biologists love it (strong biological story) but bioinformaticians struggle (too much biology, not enough computational innovation clearly articulated).

**The Solution:** Make EpiTox the protagonist, but tell its story through a compelling biological challenge. Think of it like a superhero movie: EpiTox is Spider-Man, MAGEA3 is the villain it must defeat. You need both characters to shine.

**Key Insight:** Bioinformaticians need to see COMPUTATIONAL INNOVATION clearly articulated. Right now, it's buried in details. They need to understand what's novel about EpiTox COMPUTATIONALLY, not just that it works.

## Why Bioinformaticians Are Struggling

### Issue #1: Computational Novelty Not Front and Center

#### What bioinformaticians are looking for:

- • What computational problem does EpiTox solve that others don't?
- • What algorithms/methods are novel?
- • How does it compare to existing tools COMPUTATIONALLY?
- • What are the performance metrics?

#### Current problem:

- These are scattered throughout the paper. A bioinformatician reads "EpiTox integrates multiple modules" and thinks "okay, it's a pipeline... but what's NEW?"

### Issue #2: Method Innovation Buried in Implementation Details

#### Your actual computational innovations:

- ✓ Bayesian evidence integration (NOVEL!)
- ✓ Multi-modal ranking that captures different risk types (NOVEL!)
- ✓ Population-specific SNP integration at scale (NOVEL!)
- ✓ Anchor-aware biophysical scoring (NOVEL!)

#### But these are hidden in:

- • Dense paragraphs in Methods
- • Not highlighted as innovations in Results
- • No clear comparison showing why they're better

### Issue #3: Missing the "So What?" for Bioinformaticians

- Biologists understand: "This helps us not kill patients" ✓
- Bioinformaticians need: "This solves X computational challenge in Y novel way" X

## The Solution: Dual-Narrative Structure

**Core Principle:** EpiTox is the hero. MAGEA3 is the quest. Tell both stories simultaneously.

### Think of it as Two Parallel Threads:

**Thread 1 (For Biologists):** The biological problem → EpiTox's solution → biological insights → clinical impact

**Thread 2 (For Bioinformaticians):** Computational challenges → EpiTox's innovations → performance validation → generalizable framework

**The key:** Weave these together so each audience gets what they need WITHOUT overwhelming the other.

## Proposed New Structure: The Best of Both Worlds

### INTRODUCTION: Set Up Both Narratives

#### Section 1: The Biological Problem (Current - KEEP)

- • Clinical failures with TCR-based therapies
- • MAGEA3 neurotoxicity cases
- • The cross-reactivity challenge

✓ *This works well - biologists are hooked*

#### Section 2: The Computational Challenge (ADD/EXPAND)

##### NEW CONTENT NEEDED:

- • Why existing computational methods fail:
  - - Limited to sequence similarity only
  - - Ignore population variation
  - - Single scoring metrics miss different risk types
  - - Don't integrate experimental evidence systematically
- • Frame this as: "The computational problem space"

#### Section 3: EpiTox Overview - The Hero's Introduction (REDESIGN)

**Current version:** Too brief, doesn't establish computational novelty

**New approach:** 3-4 paragraphs that clearly articulate:

##### 1. Paragraph 1: The integrative philosophy

"To address these computational and biological challenges, we developed EpiTox, a multi-modal platform that integrates sequence-based, structure-aware, and evidence-based approaches to systematically identify and prioritize pHLA off-targets..."

##### 2. Paragraph 2: Key computational innovations

"EpiTox introduces several computational advances: (1) a Bayesian framework for integrating heterogeneous experimental evidence, (2) anchor-aware biophysical scoring that weights features by functional relevance, (3) systematic population-specific variant integration, and (4) multi-modal ranking that captures complementary risk dimensions..."

##### 3. Paragraph 3: The biological implementation

"We applied EpiTox to MAGE-A3, a therapeutic target that caused fatal neurotoxicity in clinical trials, to demonstrate its capability to identify both known toxic off-targets and predict novel cross-reactive peptides..."

#### 4. Paragraph 4: Validation and impact

"Using high-throughput TCRm binding assays, we validated 538 predicted peptides, representing one of the largest experimental validation efforts for off-target prediction..."

## RESULTS: Dual Narrative in Each Section

**Strategy:** Each result section should answer BOTH:

- • What did we learn about MAGEA3? (biology)
- • How did EpiTox enable this? (computation)

### Result 1: EpiTox Architecture Enables Comprehensive Off-Target Assessment

**KEEP Figure 1 HERE - but redesign the narrative around it:**

**Opening (computational frame):**

"Systematic off-target assessment requires integrating multiple computational strategies, each addressing a different aspect of cross-reactivity risk. We designed EpiTox as a modular platform consisting of six interconnected modules (Figure 1), each tackling a specific computational challenge:"

**Then explain each module as a solution to a computational problem:**

#### X-Scan: Solving the motif characterization problem

- • Computational challenge: Determining position-specific tolerance
- • EpiTox solution: Statistical framework for classifying positions
- • MAGEA3 result: P2/P9 anchors rigid, P3/P5 permissive → explains cross-reactivity potential

#### FindTopes: Solving the search space problem

- • Computational challenge: Balancing sensitivity vs specificity in similarity search
- • EpiTox solution: Dual-algorithm approach (BLOSUM62 + PAM250)
- • MAGEA3 result: 1,481 candidates identified vs ~100 in typical searches

#### MutaTopes: Solving the population variation problem

- • Computational challenge: Integrating SNPs without combinatorial explosion
- • EpiTox solution: Genomic coordinate mapping + functional filtering
- • MAGEA3 result: 35 SNP-derived peptides with altered binding → population-specific risks

#### PrediTopes: Solving the prediction aggregation problem

- • Computational challenge: Multiple predictors, conflicting results
- • EpiTox solution: Consensus approach + experimental data integration
- • MAGEA3 result: Refined binding predictions for 1,481 peptides

#### Bayesian Module: Solving the evidence integration problem

- • Computational challenge: Heterogeneous experimental evidence with varying reliability

- • EpiTox solution: Likelihood ratio framework with context-aware weighting
- • MAGEA3 result: 286 peptides with experimental data classified into evidence tiers

#### AnnoTopes: Solving the multi-modal ranking problem

- • Computational challenge: Different risk types need different metrics
- • EpiTox solution: Bi-features (sequence+affinity) + Multi-features (biophysical) scores
- • MAGEA3 result: Complementary rankings capture different off-target categories

#### End with transition:

"Having established this computational framework, we applied EpiTox comprehensively to MAGE-A3 to demonstrate its capability to identify and prioritize physiologically relevant off-targets..."

## Result 2: Comprehensive MAGEA3 Analysis Reveals Extensive Cross-Reactivity Landscape

**Structure: Biology first, computation enables**

### Paragraph 1: The biological finding

"Application of EpiTox to MAGE-A3 (KVAELVHFL, HLA-A\*02:01) revealed an extensive cross-reactivity landscape comprising 1,481 potential off-target peptides across the human proteome. This far exceeds the two known toxic off-targets (EPS8L2 and MAGE-A12) that caused fatal neurotoxicity, suggesting substantial underestimation of safety risks by traditional methods..."

### Paragraph 2: The computational enabler

"This comprehensive identification was enabled by EpiTox's flexible search strategy. Unlike tools restricted to  $\geq 5$  amino acid identity, our dual-algorithm approach with customizable mismatch thresholds ( $\leq 4$  mismatches) captured distant but structurally relevant sequences. Position-specific motif analysis revealed why cross-reactivity is extensive: P3 and P5 tolerate 60-70% of substitutions (permissive positions), while P2 and P9 anchors remain constrained (Figure 2A)..."

### Paragraph 3: Biological implications

"Off-target distribution analysis showed that 19% harbor mismatches only in backbone positions (anchor-intact), maintaining HLA presentation while altering TCR contact surfaces. These represent high-risk candidates as they likely form stable pHLA complexes. Conversely, 31% exhibited anchor-region mismatches, potentially reducing presentation but warranting empirical validation..."

## Result 3: Population-Specific Variants Reveal Hidden Off-Target Risks

**This is your most novel contribution - emphasize the computational innovation:**

### Paragraph 1: The problem

"Current off-target prediction tools either ignore genetic variation entirely or include only common SNPs associated with disease. However, rare variants collectively account for substantial population diversity and may create or eliminate off-targets in specific individuals..."

### Paragraph 2: The computational solution

"EpiTox addresses this through systematic integration of population-specific SNPs. By mapping genomic coordinates of all identified off-target peptides and intersecting with variant databases (gnomAD, 1000 Genomes), we computationally generated SNP-derived peptides and assessed their binding potential. This approach enables identification of population-stratified risks without requiring experimental screening of every possible variant..."



### Paragraph 3: MAGEA3 findings

"For MAGE-A3, this analysis identified 35 SNP-derived peptides with altered predicted binding. Notably, Q5T2L2-FRAELVHPA (wild-type: weak binder) is converted to FKAELVHPA by rs77445156, creating a strong binder. Conversely, P21817-KLAENIHKL (rs746904839) destroys binding in the variant form. These represent population-specific safety considerations invisible to standard approaches..."

## Result 4: Multi-Modal Ranking Captures Complementary Risk Dimensions

### Computational innovation showcase:

#### Paragraph 1: The problem with single scores

"Most off-target prediction tools produce a single risk score, attempting to collapse multiple risk factors into one dimension. However, off-target risk is multi-faceted: sequence similarity indicates evolutionary relatedness, binding affinity determines presentation likelihood, and biophysical properties affect TCR recognition. These dimensions are not always correlated..."

#### Paragraph 2: EpiTox's solution

"EpiTox implements two complementary ranking strategies: Bi-features combines sequence similarity (BLOSUM62) with predicted binding affinity, capturing targets similar to MAGE-A3 that are likely presented. Multi-features computes anchor-weighted Euclidean distance across eight biophysical properties, identifying structurally similar peptides regardless of sequence. The weak correlation between these scores (Spearman  $\rho = 0.19$ ) confirms they capture orthogonal risk dimensions..."

#### Paragraph 3: Biological validation

"Disagreement analysis revealed biologically interpretable patterns: peptides ranked high by Bi-features but low by Multi-features (n=32) were sequence-similar but structurally divergent, often due to proline insertions or aromatic residue differences. Conversely, peptides ranked high by Multi-features but low by Bi-features (n=125) were structurally similar but sequence-divergent, representing the most challenging off-targets to identify by traditional methods..."

## Result 5: Experimental Validation Confirms Computational Predictions

### Balance: Show the biology AND the prediction performance:

#### Paragraph 1: Validation scope

"To empirically validate EpiTox predictions, we tested 538 peptides using high-throughput TCRm binding assays (highSCORE platform). This represents one of the largest experimental validation datasets for off-target prediction, enabling robust assessment of computational performance across multiple prediction strategies..."

#### Paragraph 2: Computational performance

"Of 538 tested peptides, 37 showed confirmed binding (7% hit rate). Binders were distributed across all prediction strategies, validating the complementary nature of our

multi-modal approach. Importantly, we identified 26 novel off-targets not previously reported in the literature or experimental databases, demonstrating EpiTox's discovery potential beyond existing knowledge..."

#### Paragraph 3: Biological insights

"Sequence logo analysis of validated binders revealed conservation patterns consistent with HLA-A\*02:01 anchor preferences (P2: L/M, P9: V/L). Gene expression profiling showed diverse tissue representation, including neuronal markers (consistent with clinical neurotoxicity) and ubiquitously expressed proteins (representing broad safety concerns). Notably, PPP2R1B-GIAELVHFS, the only peptide with mismatches exclusively at anchors, confirmed that anchor-position variation can modulate but not always eliminate binding..."

### Result 6: Bayesian Evidence Integration Prioritizes High-Confidence Off-Targets

#### Computational methodology showcase:

#### Paragraph 1: The problem

"Experimental databases contain heterogeneous evidence: some peptides are validated with specific HLA alleles in relevant tissues, while others are reported with limited context. Treating all experimental evidence equally ignores differences in reliability and relevance..."

#### Paragraph 2: Bayesian solution

"We developed a Bayesian framework that quantitatively integrates evidence quality (study count, experimental method), HLA specificity (allele match vs. class match), binding affinity, and tissue context. Likelihood ratios are assigned based on biological relevance, with multiplicative updates generating posterior probabilities that reflect confidence in each protein-peptide-HLA relationship (see Methods for full mathematical framework)..."

#### Paragraph 3: MAGEA3 application

"Of 286 MAGE-A3-related peptides with experimental evidence, our model classified 159 (11%) as high confidence (>80% posterior probability), representing the most reliable off-targets for prioritized validation. Importantly, the model distinguished sequence-identical peptides from different protein sources: MAGE-A3 and MAGE-A9 share identical peptide sequences but have different experimental evidence profiles, highlighting the importance of protein-context specificity..."

## **DISCUSSION: Address Both Audiences Explicitly**

### **Structure Proposal:**

#### **Section 1: Computational Advances (for bioinformaticians)**

- • Explicit comparison with existing tools
- • What EpiTox does that they don't (TABLE comparison)
- • Computational performance metrics
- • Scalability and runtime considerations

#### **Section 2: Biological Insights (for biologists)**

- • Implications for MAGEA3 therapeutic development
- • Why 26 novel off-targets matter
- • Population-specific risk implications
- • Lessons for other cancer-testis antigens

#### **Section 3: Methodological Considerations (both)**

- • Limitations of sequence-based approaches
- • Why multi-modal ranking is necessary
- • Integration with ML approaches (EpiPredict)

#### **Section 4: Future Directions (both)**

- • Computational: structural prediction integration
- • Biological: application to other targets

## Critical Missing Elements for Bioinformaticians

### 1. Comparison Table (ADD TO DISCUSSION)

Create a table comparing EpiTox to existing tools:

Features to compare:

- Search strategy (similarity algorithm)
- SNP integration (yes/no, which populations)
- Evidence integration (method)
- Ranking approach (single vs multi-modal)
- Peptide length flexibility
- Computational cost

**This ONE table will make bioinformaticians immediately understand what's novel.**

### 2. Performance Metrics (ADD TO RESULTS)

**Bioinformaticians need numbers:**

- Hit rate: 7% (37/538) - how does this compare to random?
- Sensitivity: How many known off-targets did you recover?
- Precision: Of high-ranked predictions, what % validated?
- Comparison: If you ran CrossDome or other tools, how do they compare?

### 3. Computational Cost Analysis (ADD TO METHODS/DISCUSSION)

- Runtime for MAGEA3 analysis
- Scalability: How long for 100 targets?
- Resource requirements

### 4. Ablation Study (Would Strengthen Paper)

**Show that each module adds value:**

- Without SNPs: X fewer off-targets identified
- Without Bayesian integration: Y% misclassified
- Single ranking only: Z disagreement cases missed

## **ABSTRACT: Redesign for Dual Audience**

**Current Abstract:** Too biology-heavy for bioinformatics journal

**Proposed Redesign (balances both):**

"Background: Off-target toxicity remains a critical barrier to development of pHLA-targeted therapeutics such as TCR-mimic antibodies. Fatal clinical outcomes in MAGE-A3 trials highlight the inadequacy of current computational prediction methods, which rely on single similarity metrics, ignore population-specific variation, and fail to systematically integrate experimental evidence.

Results: We developed EpiTox, a multi-modal computational platform that addresses these limitations through: (1) flexible sequence similarity search capturing distant but structurally relevant peptides, (2) systematic integration of population-specific SNPs, (3) Bayesian framework for quantitative evidence integration, and (4) complementary ranking strategies capturing orthogonal risk dimensions. Applied to MAGE-A3, EpiTox identified 1,481 potential off-targets, including population-specific variants undetectable by standard approaches. Experimental validation of 538 peptides confirmed 37 binders, including 26 novel off-targets not previously reported. Multi-modal ranking revealed that sequence similarity and biophysical properties capture distinct risk categories (Spearman  $\rho = 0.19$ ), demonstrating the necessity of integrative approaches.

Conclusions: EpiTox provides a comprehensive framework for systematic off-target assessment, enabling identification of diverse cross-reactive peptides including population-specific risks. The platform's modular architecture and multi-modal ranking support both targeted validation campaigns and training of machine learning models for proteome-wide off-target prediction.

Availability: Code and documentation available at [URL]. MAGEA3 dataset available at [URL]."

## **TITLE: Make It Clear This Is About the Tool**

**Current title is good but could be stronger for bioinformaticians:**

### **Current:**

"EpiTox: A Multi-Modal Computational Platform for Population-Aware Off-Target Risk Assessment in TCR-mimic therapeutics and Beyond"

### **Alternative 1 (emphasizes innovation):**

"EpiTox: Multi-Modal Framework for Population-Aware Off-Target Prediction Reveals Extensive MAGE-A3 Cross-Reactivity"

### **Alternative 2 (emphasizes both):**

"EpiTox: Integrated Computational Platform Combining Sequence, Structure, and Evidence-Based Approaches for Systematic pHLA Off-Target Assessment"

**Recommendation:** Keep current title - it's good. Just ensure the paper delivers on the "multi-modal" and "population-aware" promises prominently.

## Writing Tips: Speaking to Both Audiences

### For Bioinformaticians, Add These Phrases:

- "This computational challenge requires..."
- "Our algorithm addresses this by..."
- "Compared to existing methods that..."
- "Performance metrics show..."
- "Runtime scales as  $O(n)$ ..."

### For Biologists, Keep These Phrases:

- "Clinical trials revealed..."
- "These findings suggest..."
- "Tissue expression analysis shows..."
- "This has implications for..."

### Universal Structure:

- Problem → EpiTox's computational solution → Biological finding → Implication

### Example:

"[PROBLEM] Population-specific genetic variation creates patient-specific off-target risks, but systematic integration of SNPs is computationally prohibitive. [SOLUTION] EpiTox addresses this through genomic coordinate mapping and functional filtering, reducing the search space from billions of theoretical variants to experimentally tractable sets. [FINDING] For MAGE-A3, this identified 35 SNP-derived peptides with altered binding, including rs77445156 which creates a strong binder from a weak binder. [IMPLICATION] This represents a population-stratified risk requiring ethnic consideration in patient selection."

## FIGURES: Redesign to Serve Both Audiences

### Figure 1: EpiTox Architecture (KEEP IN MAIN, REDESIGN)

**Current weakness:** Shows modules but not the computational innovation

**Redesign to include:**

- Panel A: Modular architecture (current)
- Panel B: Computational challenge each module solves
- Panel C: MAGEA3 workflow example showing data flow

**Legend should emphasize:**

- "EpiTox addresses six computational challenges in off-target prediction through modular, integrative design..."

### Figure 2: MAGEA3 Landscape (CURRENT IS GOOD)

**Minor addition:**

- • Add a panel comparing EpiTox search space vs. typical tools
- • Show: EpiTox found 1,481 vs. typical ~100-200

### NEW Figure: Computational Performance

**Add a figure showing:**

- Panel A: Hit rate across different ranking strategies
- Panel B: Precision-recall curves if available
- Panel C: Comparison with other tools (if you have data)

### Figure 4: Validation (KEEP, MAYBE ADD)

**Consider adding:**

- Panel showing which ranking method caught which binders
- Demonstrates the value of multi-modal approach



## Final Checklist: Balancing Both Audiences

### For Biologists (Already Strong):

- ✓ Clinical context clear
- ✓ MAGEA3 story compelling
- ✓ Safety implications evident
- ✓ Experimental validation robust

### For Bioinformaticians (Needs Strengthening):

- ☐ Computational innovations clearly articulated in Introduction
- ☐ Each Result section starts with "computational challenge + solution"
- ☐ Comparison table with existing tools in Discussion
- ☐ Performance metrics reported
- ☐ Computational cost discussed
- ☐ Ablation study showing module contributions

### Structure Check:

- ☐ Introduction sets up BOTH biological and computational challenges
- ☐ EpiTox introduced as solution to both
- ☐ Results show "Challenge → EpiTox solution → Finding → Implication"
- ☐ Discussion addresses both audiences explicitly
- ☐ Figures serve both audiences

## The Key Insight: What Makes a Good Tool Paper

**A great tool paper needs:**

5. 1. Clear problem statement (computational + biological)
6. 2. Novel methodological contributions (algorithms, frameworks)
7. 3. Demonstration on real problem (MAGEA3)
8. 4. Biological insights that couldn't be obtained otherwise
9. 5. Performance validation against existing methods
10. 6. Generalizable framework (not just for MAGEA3)

**You have 1, 3, 4, and 6. You need to strengthen 2 and 5.**

### Specifically:

#### **#2 - Novel methodological contributions:**

- Make these EXPLICIT in Introduction and early Results
- Don't assume readers will figure out what's novel
- State clearly: "Unlike existing methods that X, EpiTox does Y"

#### **#5 - Performance validation:**

- Add comparison with other tools (even if limited)
- Quantify performance improvements
- Show ablation study demonstrating each module's value

## Action Plan: Priority Edits

### CRITICAL (Do First):

11. 1. Expand Introduction Section 2: Add "Computational Challenges" (2 paragraphs)
12. 2. Expand Introduction Section 3: Redesign EpiTox overview (3-4 paragraphs)
13. 3. Redesign Result 1: Frame each module as solving a computational problem
14. 4. Add comparison table to Discussion

### HIGH PRIORITY (Do Next):

15. 5. Add performance metrics to Result 5 (validation)
16. 6. Restructure each Result section: Challenge → Solution → Finding → Implication
17. 7. Redesign Figure 1 to show computational innovations
18. 8. Revise Abstract to balance both audiences

### MEDIUM PRIORITY (Polish Phase):

19. 9. Add computational cost analysis to Methods/Discussion
20. 10. Expand Discussion Section 1 (Computational Advances)
21. 11. Add "what's novel" statements throughout

## Example: Introduction Section 2 (NEW)

### The Computational Challenges of Off-Target Prediction

Despite advances in peptide-HLA binding prediction, systematic identification of physiologically relevant off-targets remains computationally challenging. First, the sequence similarity search space is vast: even restricting to 9-mer peptides, the human proteome contains millions of potential epitopes, requiring efficient search strategies that balance sensitivity and specificity. Second, population-specific genetic variation creates patient-specific off-target landscapes, but systematic integration of SNPs leads to combinatorial explosion without intelligent filtering. Third, experimental evidence for peptide-HLA interactions exists in fragmented form across multiple databases with varying levels of reliability, lacking a principled framework for quantitative integration. Fourth, off-target risk is multi-dimensional: sequence similarity indicates evolutionary relatedness, binding affinity determines presentation likelihood, and biophysical properties affect TCR recognition. These dimensions are poorly correlated, yet most tools attempt to reduce risk to a single score, potentially missing important off-target categories.

Existing computational tools address subsets of these challenges but lack comprehensive integration. Sequence-based tools like Expitope and iCrossR rely on amino acid identity and HLA binding predictions but ignore population variation and biophysical properties. Biochemical similarity frameworks such as dGraph and sCRAP leverage physicochemical descriptors but lack systematic evidence integration. CrossDome represents a significant advance through integration of experimental immunopeptidomics data, but its search space is limited to pre-validated peptides, constraining discovery of novel off-targets. ARDiTox incorporates some SNPs but requires high sequence similarity ( $\geq 5$  identical amino acids) and focuses on disease-associated mutations rather than population-wide variation. No existing tool systematically addresses all four computational challenges in an integrated framework.

## Example: Result Section 3 Rewrite (Population SNPs)

### Result 3: Population-Specific Genetic Variation Reveals Hidden Off-Target Risks

#### The Computational Challenge:

Single nucleotide polymorphisms (SNPs) can create or eliminate off-target peptides in specific populations, yet systematic integration is computationally prohibitive. The human genome contains >650 million SNPs (gnomAD), and each protein-coding SNP potentially generates multiple altered peptides depending on its position within potential epitope windows. Combinatorial enumeration would generate billions of theoretical variants, the vast majority functionally irrelevant.

#### EpiTox's Solution - MutaTopes Module:

We developed a coordinate-based mapping strategy that dramatically reduces the search space. First, FindTopes identifies all potential off-target peptides and extracts their genomic coordinates. Second, MutaTopes intersects these coordinates with population-specific SNP databases, filtering to non-synonymous variants within epitope windows. Third, variant peptides are generated and assessed for HLA binding potential. This targeted approach focuses computational resources on functionally plausible variants, reducing billions of theoretical combinations to tractable experimental sets while maintaining comprehensive coverage of population-relevant variation.

#### MAGEA3 Findings:

For MAGE-A3, systematic SNP integration identified 35 variant-derived peptides representing population-specific off-target risks (Table 2, Figure X). Three cases exemplify the biological significance:

Q5T2L2-FRAELVHPA (rs77445156, c.2512T>A, p.Phe838Lys): The wildtype peptide shows moderate HLA-A\*02:01 affinity (KD = 1,560 nM). However, rs77445156 (MAF = 0.18% in European populations) creates FKAELVHPA with significantly enhanced binding (KD = 180 nM), representing a ~8-fold increase in predicted affinity. Experimental validation confirmed strong TCRm binding for the variant but not wildtype peptide. This SNP, present in ~1 in 550 individuals of European ancestry, creates a high-risk off-target invisible to reference genome analysis.

P21817-KLAENIHKL (rs746904839, p.Glu879Lys): Represents the opposite scenario. The wildtype peptide ELAENIHKL shows strong predicted binding (KD = 45 nM), but the variant eliminates the anchor glutamate, abolishing presentation (KD > 5,000 nM). Carriers of this

variant (MAF = 0.02%) would lack this particular off-target, illustrating how SNPs can modulate individual safety profiles bidirectionally.

A6NJW4-IVAYLVHYV (rs145254459, p.Ile167Val): The SNP affects a non-anchor position (P1), generating VVAYLVHYV. Both wildtype and variant show similar predicted binding ( $K_D$  = 230 vs 310 nM), but the I→V substitution alters surface properties potentially affecting TCR recognition despite preserved HLA binding. This exemplifies subtle structural changes detectable by multi-biophysical scoring but invisible to affinity-only approaches.

### **Biological Implications:**

These findings demonstrate that population genetic variation creates patient-specific off-target landscapes requiring consideration in therapeutic development. For MAGE-A3, ~0.2% of European-ancestry individuals harbor SNPs creating novel high-risk off-targets, while others possess variants eliminating predicted off-targets. This suggests that comprehensive safety assessment may require population-stratified risk analysis or, ultimately, personalized off-target profiling for high-risk therapeutics.

## Summary: Making EpiTox Shine for Both Audiences

### The bottom line:

Your paper has:

- ✓ Excellent science
- ✓ Strong biological story
- ✓ Robust experimental validation
- ✓ Novel computational methods

What it needs:

- • Explicit articulation of computational innovations
- • Clear framing of computational challenges solved
- • Performance metrics and comparisons
- • Better balance between describing what EpiTox does vs. what it found

### The Fix:

- Make each section answer BOTH questions:
  - 1. What computational problem did we solve?
  - 2. What biological insight did this enable?

**This way EpiTox remains the main character, but both audiences can appreciate why it's a hero.**